DSP – LEC – 2 - Ch-1

CHAPTER – 1

ANALOG AND DIGITAL SIGNAL PROCESSING

Ch. 1- index

ANALOG AND DIGITAL SIGNAL PROCESSING

1.1 SIGNALS, SYSTEMS, AND SIGNAL PROCESSING

1.2 Basic Elements of analog and digital signal processing systems

1.3 Advantages of Digital V.S Analog Sig. Processing

Ch. 1- index -1

1.4 Classification of Signals

1.4.1 Continuous-Time versus Discrete-Time Signals

1.4.2 Continuous-Valued V.S Discrete-Valued Signals

1.4.3 Deterministic V.S Random Signals

1.4.4 Multichannel and Multidimensional Signals

Sec. 1.1

1.1 SIGNALS, SYSTEMS, AND SIGNAL PROCESSING

SIGNALS

- A signal is defined as any physical quantity that varies with time, space, or any other independent variable (s).

- Mathematically, we describe a signal as a function of one or more independent variables.

Sec. 1.1 Cont. - 1

-For example, the functions

$$s1(t) = 5t$$

$$s2(t) = 20t2 \qquad (1.1)$$

describe two signals, one that varies linearly with the independent variable time and second that varies quadratically with " t ".

-Equation (1.2), describes a signal of two independent variables x and y that could represent the two spatial coordinates in a plane.

$$s(x, y) = 3x + 2xy + 10y2 \qquad (1.2)$$

Sec. 1.1 Cont. - 2

-**For example,** as speech signal cannot be described functionally by expressions such as (1.1).

-**In general,** a segment of speech signal may be represented to a high degree of accuracy as a sum of several sinusoids of different amplitudes and frequencies, as (1.3)

$$\sum_{i=1}^{N} A_i(t) \sin\left[2\pi f_i(t)t + \theta_i(t)\right] \qquad (1.3)$$

Where: {Ai(t)}, fi(t), and {θi (t)} are the sets of possibly time- varying amplitudes, frequencies, and phases, respectively, of the sinusoids.

Sec. 1.1 Cont. - 3

SYSTEM

-It is defined as a physical device that performs an operation on a signal.

- For example, a filter used to reduce the noise and interference corrupting a desired information is called a system.

-In this case the filter performs some operation(s) on the signal, which has the effect of reducing (filtering) the noise and interference for the desired information.

Sec. 1.1 Cont. - 4

SIGNAL PROCESSING

- **When we pass a signal through a system, as in filtering, we say that we have processed the signal.**

- **In this case the processing of the signal involves filtering the noise and interference from the desired signal.**

Sec. 1.1 Cont. -5

- **In general, the system is characterized by the type of operation that it performs on the signal.**

- **For example, if the operation is linear, the system is called linear. If the operation on the signal is nonlinear, the system is said to be nonlinear, and so forth.**

Sec. 1.1 Cont. - 6

- **Alternatively, the digital processing on the signal may be performed by digital hardware (logic circuits) configured to perform the desired specified operations.**

- **In such a realization, we have a physical device that performs the specified operations.**

- **In a broader sense, a digital system can be implemented as a combination of digital hardware and software, each of which performs its own set of specified operations.**

Sec. 1.2

1.2 Basic Elements of analog and dig. signal processing systems

Analog Signal Processors

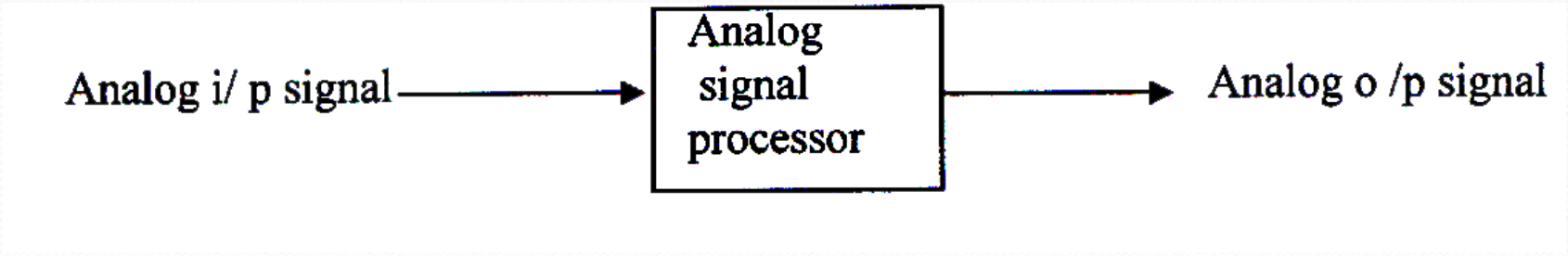


Fig. 1.1 Block diagram of analog signal processor.

Sec. 1.2 Cont. - 1

Digital Signal Processors

- **Digital signal processing provides an alternative method for processing the analog signal, as illustrated in Fig. 1.2.**

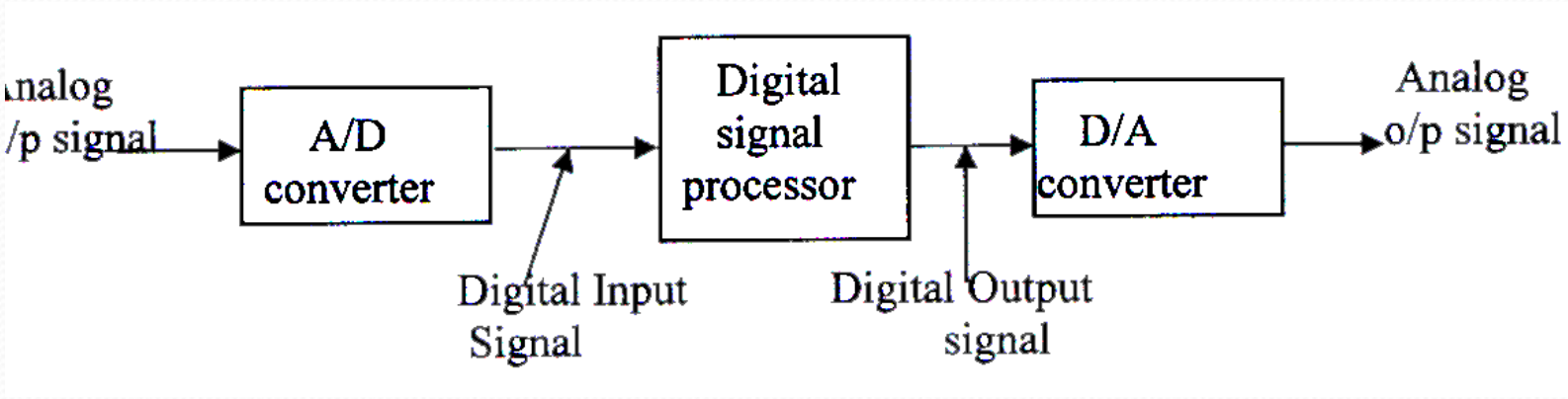


Fig. 1.2 Block diagram of a digital signal processor.

Sec. 1.3 Cont. - 1

1.3 Advantages of Dig. Sig. Processing V.S Analog Sig. Processing

(1) Flexibility: A digital programmable system allows flexibility in reconfiguring the digital signal processing operations simply by changing the program.

- Reconfiguration of an analog system usually implies a redesign of the hardware followed by testing and verification to see that it operates properly.

Sec. 1.3 Cont. - 2

(2) Accuracy considerations also play an important role in determining the form of the signal processor.

- **Tolerances in analog circuit components make it extremely difficult for the system designer to control the accuracy of an analog signal processing system.**

- **On the other hand, a digital system provides much better control of accuracy requirements.**

Sec. 1.3 Cont. - 3

(3) Digital signals are easily stored on magnetic media (tape or disk) without deterioration or loss of signal fidelity beyond that introduced in the A/D conversion.

As a consequence, the signals become transportable and can be processed off-line in a remote laboratory.

(4) The digital signal processing method also allows for the implementation of more sophisticated signal processing algorithms.

Sec. 1.3 Cont. - 4

(5) It is usually very difficult to perform precise mathematical operations on signals in analog form but these same operations can be routinely implemented on a digital computer using software.

(6) In some cases a digital implementation of the signal processing system is cheaper than its analog counterpart.

- **The lower cost may be due to the fact that the digital hardware is cheaper, or perhaps it is a result of the flexibility for modifications provided by the digital implementation.**

Sec. 1.3 Cont. - 5

Digital implementation limitations

- One practical limitation is the speed of operation of A/D converters and digital signal processors.

- So, the signals having extremely wide bandwidths require fast-sampling-rate A/D converters and fast digital signal processors.

- Hence there are analog signals with large bandwidths for which a digital processing approach is beyond the state of the art of digital hardware.

Sec. 1.4

1.4 Classification of Signals

1.4.1 Continuous-Time versus Discrete-Time Signals

1.4.2 Continuous-Valued V.S Discrete-Valued Signals

1.4.3 Deterministic V.S Random Signals

1.4.4 Multichannel and Multidimensional Signals

Sec. 1.4 Cont. - 1

1.4.1 Continuous-Time versus Discrete-Time Signals

- Continuous-time signals or analog signals are defined for every value of time and they take on values in the continuous interval (a, b), where "a" can be "-∞ "and "b" can be " ∞ ".

- Mathematically, these signals can be described by functions of a continuous variable.
- The signals $x_1(t) = \cos(t)$, $x_2(t) = e^{-|t|}$, $-\infty < t < \infty$ are example of analog signals.

Sec. 1.4 Cont. - 2

- Discrete-time signals are defined only at certain specific values of time. These time instants need not be equidistant, but in practice they are usually taken at equally spaced intervals for computational convenience and mathematical tractability.

- The signal $x(t_n)= e^{-|t_n|}$, n = 0 , 1, 2, . . provides an example of a discrete-time signal.

Sec. 1.4 Cont. - 3

- If the time instants t_n are equally spaced (i.e., $t_n = nT$), the notation x(nT) is also used. For example, the sequence .

$$X(n) = \begin{cases} 0.8^n, & \text{if } n \geq 0 \\ 0, & \text{otherwise} \end{cases} \qquad (1.4)$$

is a discrete-time signal, which is represented graphically as in Fig. 1.3

Sec. 1.4 Cont. - 4

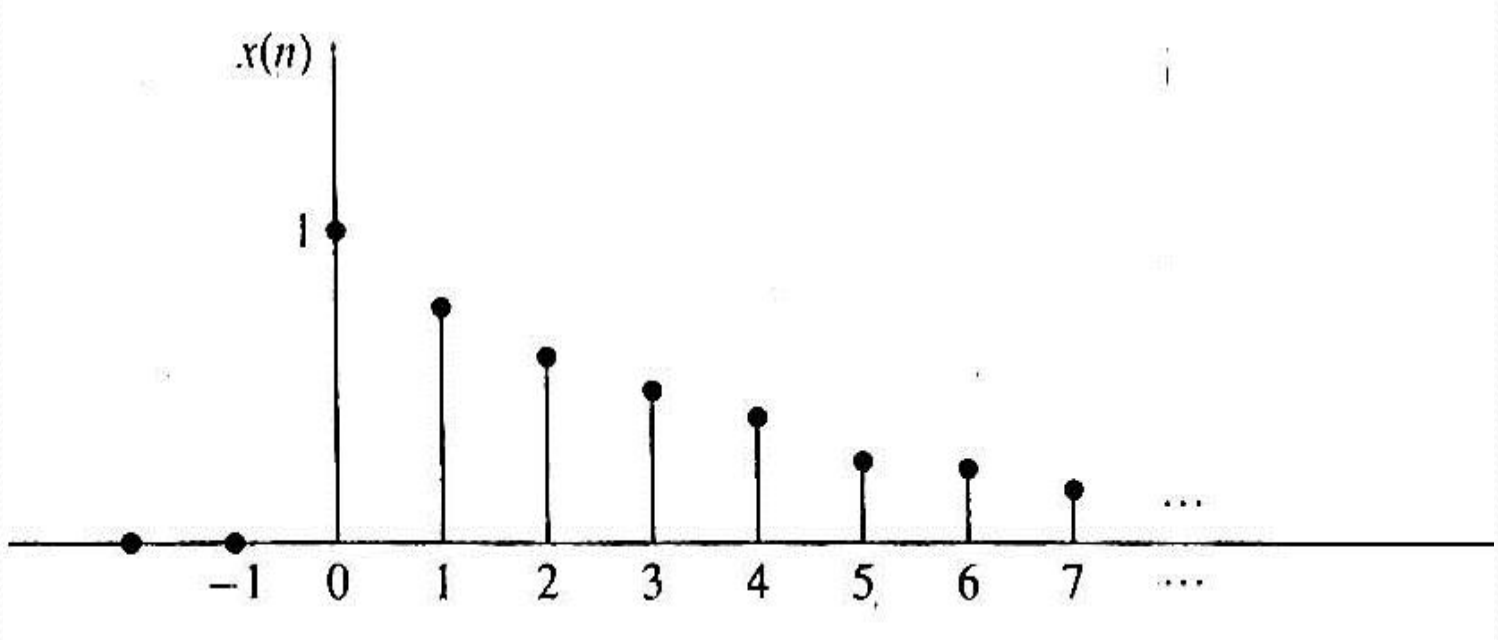


Fig 1.3 DIscrete time signal:

$$X(n) = \begin{cases} 0.8^n, & \text{if } n \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

Sec. 1.4 Cont. - 5

1.4.2 Continuous Valued V.S Disc. Valued Sig.s

- The values of a continuous-time or discrete-time signal can be continuous or discrete. If a signal takes on all possible values on a finite or an infinite range, it is to be continuous-valued signal.

- Alternatively, if the signal takes on values from a finite set of possible values, it is said to be a discrete-valued signal.

- Usually, these values are equidistant and hence can be expressed as an integer multiple of the distance between two successive values.

Sec. 1.4 Cont. - 6

- **Figure 1.4 shows a digital signal that takes on one of four possible values. In order for a signal to be processed digitally, it must be discrete in time and its values must be discrete (i.e., it must be a digital signal).**

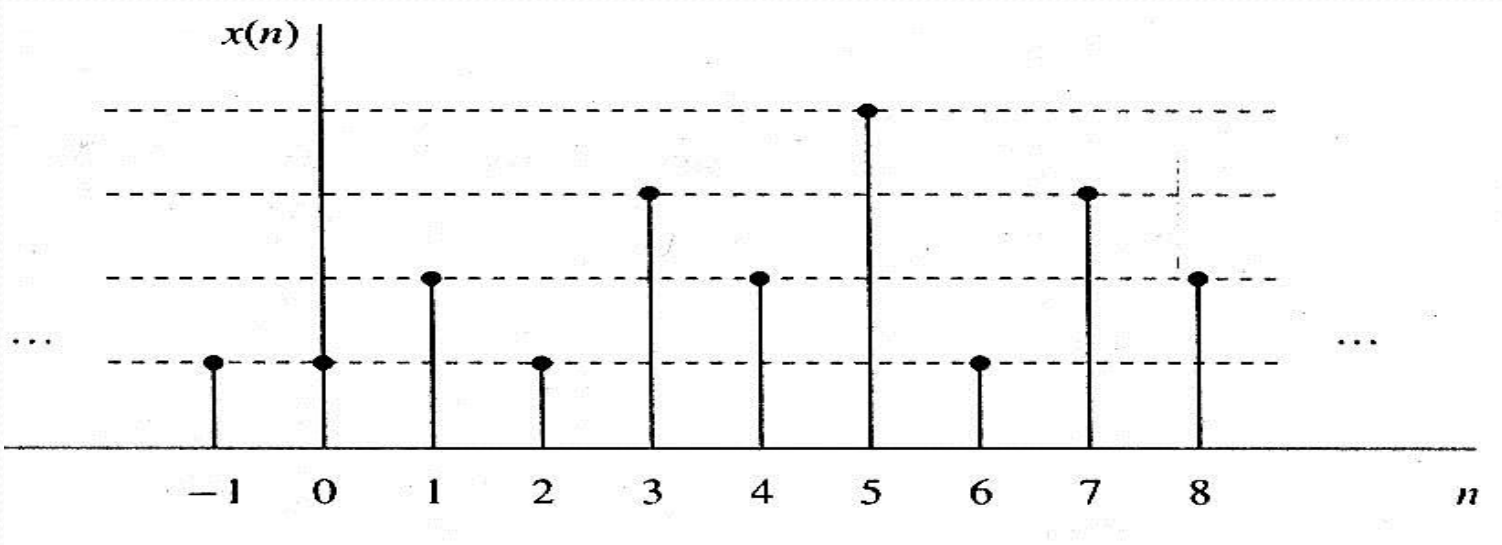


Fig. 1.4 Digital signal with four different amplitude values.

Sec. 1.4 Cont. - 7

- **The process of converting a continuous-valued signal into a discrete-valued signal, called quantization, is basically an approximation process. It may be accomplished simply by rounding or truncation.**

- **For example, if the allowable signal values in the digital signal are integers, say 0 through 15, the continuous-value signal is quantized into these integer values.**

Thus the signal value 8.58 will be approximated by the value 8 if the quantization process is performed by truncation or by 9 if the quantization process is performed by rounding to the nearest integer.

Sec. 1.4 Cont. - 8

1.4.3 Deterministic Versus Random Signals

- Any signal that can be uniquely described by an explicit mathematical expression, a table of data, or a well-defined rule is called deterministic.

- This term is used to emphasize the fact that all past, present, and future values of the signal are known precisely, without any uncertainty.

- Otherwise, we refer to these signals as random.

Sec. 1.4 Cont. - 9

1.4.4 Multichannel and Multidimensional Signals

- As explained in section 1.1, a signal is described by a function of one or more independent variables. The value of the function (i.e., the dependent variable) can be a real-valued scalar quantity, a complex-valued quantity, or perhaps a vector.

- E,g., the signal $s1(t) = A \sin 3\pi t$ is a real-valued signal.

- However, the signal $s2(t) = Aej3\pi t = A \cos 3\pi t + j A \sin 3\pi t$ is complex valued signal.

Sec. 1.4 Cont. - 10

- In some applications, signals are generated by multiple sources of multiple sensors. Such signals, in turn, can be represented in vector form.
- If , k = 1,2,3, denotes the electrical signal from the kth sensor as a function of time, the set of k = 3 signals can be represented by a vector S3 (t), where

$$S3(t) = \begin{bmatrix} s_1(t) \\ s_2(t) \\ s_3(t) \end{bmatrix}$$

we refer to such a vector of signals as a multichannel signal.

Sec. 1.4 Cont. - 11

- Let us now turn our attention to the independent variable (l).

- If the signal is a function of a single independent variable, the signal is called a one-dimensional signal.

- On the other hand, a signal is called **M-dimensional** if its value is a function of **M independent variables**.

- In our Course, we deal mainly with **single-channel; one-dimensional real-or complex-valued signals** and we refer to them simply as signals.